

Handoff: everything you need for the deck

For: whoever is building the Final Capstone slides

From: the Scope v2 implementation work, 2026-08-15

Status of the code: phases 0 to 12 done except the deck itself

You own the deck. This document gives you the numbers, the story behind them, and the things you should not claim. Everything here is measured unless it says otherwise, and where a figure is an assumption it says so, because the specification grades that distinction under value proposition and units.

1. First, the housekeeping

The work is on a branch, not on main. Everything below lives on `scope-v2-build`. `main` still has only the dataset commits. Someone needs to open the pull request:

```
gh pr create --base main --head scope-v2-build --repo Dawsxn/DrMundo
```

I did not force-push over `main` and did not touch your two commits.

The media is in git history and that is going to hurt. `labrequests/media/` is about 380 MB of PNGs, now permanent unless the history is rewritten with `filter-repo` or `BFG`. Every clone pays for it, and a fetch of that repo took over ten minutes on a normal connection. The images regenerate from master seed 20260814, so committing them bought very little. Rewriting gets more painful the longer it waits, and everyone has to re-clone when it happens. Worth deciding soon rather than after more commits land on top.

Two servers, if you want to demo live. `uvicorn api.main:app` on port 8000 and `streamlit run ui/app.py` on 8501. Both were verified working end to end with the real model, not just in tests.

2. The three headline numbers

The specification asks for at least three quantitative metrics with interpretation. These are the ones to lead with.

Metric	Result	What it means
Reading F1 (GPT-4.1, n=40)	77.2%	Recall 67.2%, precision 90.8%. The system misses tests more often than it invents them
Top-1 match accuracy	91.3%	Of the labels read, this fraction reaches a real priced MMC catalogue row
Hallucinated-peso rate	0.0%	Across 40 generated reports, no figure appeared that could not be traced to structure

The third one is the project's whole thesis, so give it a slide of its own rather than a row in a table. It is the claim that separates this from a chatbot guessing prices.

Two supporting numbers worth having ready for questions:

- **Resolution, label to code: 100%** (1452 of 1452). With reading held perfect, nothing is

lost turning a printed label into a catalogue code. This bounds the whole pipeline: no reader improvement can be undone by the resolver.

- **Latency: mean 3.6 s, max 7.2 s** against a 20 second budget that was fixed in writing

before any benchmark ran. If someone asks how the budget was chosen, that ordering is the answer.

3. The RRL story, which is the part with the most marks in it

Component 14 requires a CV or DS model, and section 6.3 requires you to present the review that led to choosing it. There are two findings and they are both real.

Finding one: OCR structurally cannot do this task

We ran docTR against a clean form. It read 186 words correctly, which is every printed label on the page, including the roughly forty tests the doctor did not order. The ground truth for that form is seven tests.

The point to make on the slide: OCR answers "what text is on this page". We need "which rows are marked", and a mark is not text. Making the OCR family competitive needs a second stage that detects checkboxes and measures ink density. That is buildable and it is a different project.

This is a better RRL slide than a table of accuracy numbers, because it explains *why* one family was rejected rather than just reporting that it scored lower.

Finding two: model capability is a threshold, not a gradient

Same image, same prompt, three models:

Model	Tests reported	Invented	Latency
gpt-4o-mini	24	18	17.5 s
gpt-4o	6	0	7.0 s
gpt-4.1	6	0	1.9 s

Ground truth was seven. GPT-4o-mini did not score slightly worse. It failed in exactly the way OCR does, listing every label on the page. No prompt work moved it.

Worth saying out loud: gpt-4o-mini is the model the rest of the application uses for chat. The vision reader deliberately does not inherit that setting. That is a design decision with evidence behind it, which is the kind of thing the rubric rewards.

The degradation curve

The evaluation set renders each form in five conditions. This is the graph to put on the slide.

Media	Recall	Invented
clean	81.7%	3
scan	74.1%	3
photocopy, 1st generation	62.5%	3
photocopy, 3rd generation	61.4%	6
fax	50.0%	2

Half the tests on a faxed form are missed. Do not hide this. It is the clearest statement of where the system's limit is, and stating it yourself is much stronger than being asked.

4. Things you should NOT claim

These will get caught in questions if you overstate them.

Do not say it works on real doctor's requests. Every image in the evaluation set is synthetic. There is no real-slip arm at all. The numbers describe generated forms, not a phone photograph of a real request in bad light.

Do not say the full waterfall runs from an image. Every form in the dataset is a laboratory request and none names a procedure, so the PhilHealth leg never fires from the image path. It fires from the conversation. The thesis is honestly stated as "images plus one question", not "images alone".

Do not present the HMO figures as anyone's actual policy. They come from Maxicare's published individual and family tiers. Most Philippine coverage is employer-negotiated and will match no public tier. The app labels this on screen and the deck should too.

Do not claim cancelled-row handling is perfect. One cancelled row in eleven was billed, because the vision model misread a strike-through. The pricing logic never bills a cancelled row, and that is tested, but the reader can still mislabel one. This is the failure that takes money from a patient, so name it as the top open issue.

Do not quote clean-form recall as 100%. An early three-image spot check gave that. The larger 40-form run gave 81.7%. Use the larger number.

5. The sanity check the specification asks for

Section 5 asks whether ChatGPT or a Google search would suffice. The answer is no and it is provable, which makes a good slide:

- MMC's price list only returns data to a POST search. Fetched plainly, the page is empty.

A general chatbot cannot see these prices.

- The mapping from MMC's catalogue names to PhilHealth RVS codes does not exist in public.

Ours is 51 mappings made by hand, and a wrong one moves an estimate by tens of thousands of pesos.

- Reading a slip requires deciding which rows carry a mark and which are struck through.

That is not a retrieval problem.

Also pre-empt this one, because a grader may raise it: the rubric lists "safety-critical workflows without a human in the loop" as a poor fit for agentic AI. Dr. Mundo estimates prices. It does not triage, diagnose, or recommend treatment, and the input guardrail refuses medical advice outright. Say that on a slide rather than in an answer.

6. Value proposition and units

The specification is specific about qualifying units, so keep the three categories separate.

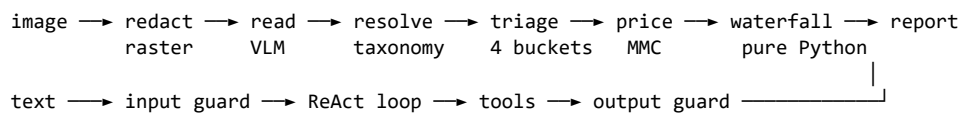
Measured. Every laboratory panel MMC publishes costs less than buying its component tests separately, by ₱60 to ₱1,400 depending on the panel and which end of the price range applies. This comes from the committed dataset and has a test behind it.

Assumed, and label it as an assumption on the slide. Pricing six tests by telephone takes roughly 30 minutes, allowing for transfers between departments. We did not measure this. Dr. Mundo answers in a mean of 3.6 seconds.

Intangible, and label it as intangible. Avoided surprise at admitting. Being told your bill is twice what you brought is a different kind of harm from losing half an hour, and we declined to invent a peso figure for it. The specification explicitly allows intangibles if you flag them, so flag it.

7. Architecture slide

The specification wants a diagram showing where the CV model plugs into the agent pipeline. This is the shape:



Three things worth calling out on that slide:

- 1 The text path is the midterm system, unchanged. This is an extension, not a rewrite.
- 2 The waterfall is pure Python with no model involvement, which is why every peso is

auditable.

- 1 Both paths converge at the output guardrail, which is what produces the 0% hallucinated

figure.

8. Component ownership

Team of three needs six components, each member owning at least two. The build assigns nine.

Member	Components	Where in the code
A	14 CV/DS, 2 Disambiguation, 5 Guardrails	vision/, guardrails/pii.py, needs-confirmation routing
B	3 RAG, 12 Advanced RAG, 10 SQL/Planning	db/, pricing/
C	9 ReAct/Tool use, 13 Evals, 8 LLMOps	agent/, eval/, MLflow
team	1 Prompting, 4 Memory, 6 UI, 7 API	carried from the midterm

Adjust the names to who actually did what. Everyone has to be able to explain their own components in questions, so pick the split that matches reality rather than the one that looks tidiest.

9. Demo

Use laparoscopic cholecystectomy or lipid profile. Not appendectomy: MMC publishes professional fees for it but no operating-room package, so it produces an almost empty answer. The midterm demo was built on appendectomy and no longer works well.

A good live sequence:

- 1 Upload a request slip. The report appears immediately, labelled as being before any HMO.

Point out that it asked nothing first.

- 1 Answer one refine question, for example the HMO plan and remaining balance. The number

drops. Point out there was no second upload.

- 1 Show the buckets: priced, not priced, please confirm, crossed out. The "not priced" bucket

is the honesty story, and "crossed out" is the one that saves the patient money.

Have a fallback recording. The specification says a failed live demo affects the presentation score and that you should disclose a fallback upfront. The reading stage makes a network call, so a bad connection is a real risk.

10. Where the numbers live

Everything above can be regenerated:

```
python -m eval.suite --offline          # deterministic layers, free
python -m eval.suite --limit 40         # includes the vision model, costs API calls
python -m eval.resolver_bench          # resolution and the section-heading ablation
python -m eval.reader_bench --limit 25 # the RRL table
```

Raw results are committed at `eval/results_v2.json` and `eval/rri_gpt41.json`. The full technical write-up is `TECHNICAL_WRITEUP.md` and `TECHNICAL_WRITEUP.pdf`, and it covers the same ground in more depth if you need wording for a slide.

The design decisions and the reasoning behind them are in `PLAN_SCOPE_V2_BUILD.md`, section 2 in particular. If a grader asks why something was built a certain way, the answer is usually written down there.